Devasheesh Mishra

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Skills & Interest

Programming Languages: Python, C/C++ and JS/TS
Frameworks & Libraries: PyTorch, Transformers, Pydantic, FastAPI, Pandas, NumPy, Matplotlib
Cloud & DevOps: Azure, Docker, Kubernates (k3s)
Infrastructure: Portainer, Traefik, Harbor, Terraform
CI/CD: GitLab CI, GitHub Actions
Databases: PostgreSQL, MongoDB, Milvus, Minio, Redis, Neo4j
Message Queues/Streaming: Kafka, Rabbit MQ
Testing: Pytest (Unit Testing and Integration Testing), Selenium
Monitoring & Logging: Prometheus, Grafana, Loki, Promtail, Alert Manager
Embedded Systems: ESP IDF, PlatformIO, Arduino, ESP32/ESP8266, ESP32-CAM, NodeMCU, Raspberry Pi
Others: Maybe I can't remember them right now.

Work Experience

Founder and CEO — January 2025 – Present
Recallr AI Inc. — *San Francisco, USA — On-site*

- Built Recallr, a persistent, queryable long-term memory layer for conversational AI systems that retains facts, preferences, relationships, and decisions across multiple conversations while preserving the source of each memory. Architected its ingestion pipeline, evolving knowledge graph, semantic retrieval, temporal reasoning, knowledge-update handling, and configurable merge and conflict-resolution rules.
- Evaluated Recallr on the LongMemEval benchmark, achieving 97.5% overall accuracy, including 97.0% temporal-reasoning accuracy and 97.4% knowledge-update accuracy. Delivered P95 latency of 408 ms for Low-Latency recall, 1.575 seconds for Balanced recall, and 8.619 seconds for Agentic recall.
- Evolved Recallr into the intelligence and memory layer for private capital, transforming fragmented deal memos, data rooms, diligence, meeting transcripts, partner notes, filings, returns, and investment-committee history into a continuously updated, queryable decision graph that preserves how a firm's investment judgment evolves over time.

Canopy 2026 — March 2026 – May 2026
Founders, Inc. — *San Francisco, California, United States — On-site*

- Selected as one of 100 teams globally for Canopy 2026, a five-week on-site builder residency at Founders, Inc.'s San Francisco campus.
- Built and commercialized Recallr's memory layer for conversational AI systems during the program, acquiring multiple clients and generating approximately $15,000 in revenue.

AI/ML Engineer — September 2025 – November 2025
MIRA — *San Francisco, California, United States — Remote*

- Real-Time Streaming Speech-to-Text Pipeline: Designed and implemented a low-latency, WebSocket-based streaming transcription system using Soniox STT models (stt-rt-preview-v2), with real-time speaker diarization, silence detection, and multilingual translation. Built a comprehensive benchmarking framework to evaluate ASR performance across Soniox, Deepgram, Google, and AssemblyAI using metrics such as word error rate, latency, and speaker attribution accuracy. Used empirical results to guide model selection and system design decisions.
- Voice Fingerprinting and Speaker Verification: Engineered an end-to-end voice biometric pipeline covering user enrollment, audio signal serialization, PostgreSQL storage, and real-time speaker identification. Used SpeechBrain's ECAPA-TDNN model trained on VoxCeleb to generate and compare speaker embeddings. Deployed the system as a standalone FastAPI microservice on AWS ECS, with parallel cosine-similarity verification across diarized speaker segments, enabling identification of the user's voice against ambient speakers with under 200 ms inference latency.
- Long-Term Memory Extraction from Personal Data: Built an LLM-powered memory extraction pipeline using Gemini 2.0 Flash to process a user's Gmail corpus and identify persistent signals such as personality traits, areas of expertise, preferences, and behavioral patterns. Structured the extracted information into long-term user knowledge representations, enabling AI agents to maintain contextual memory beyond individual conversations.

Founder and CEO — May 2024 – January 2025
Stapes AI — *Delhi, India — On-site*

- Founded an AI Home Automation company, successfully shipping v1.0 of the product in a month and acquired 50+ beta testers.

- Developed a high-end home automation solution with integrations for Smart TVs, Fire Sticks, and switch boards, enabling control via voice and mobile app. Planned to develop a circular custom device like alexa echo show with a 7-inch display for a complete home automation experience.
- Selected for buildspace's Nights & Weekends S5 program, a startup accelerator backed by Y Combinator and a16z.
- Now whitelabeling the home automation tech to other players. Planning to restart the company with more focus on AI after aquisition.

AI/ML Engineer Intern August 2024 – September 2024
Proeffico Solutions Private Ltd. *Noida, India — On-site*

- Developed "RDBMS Chat", an internal AI tool enabling non-technical staff to query complex databases using natural language, enhancing data accessibility and decision-making processes.
- Provided IT infrastructure support for servers and VMs, ensuring operational stability for various backend systems.

Builder, Nights & Weekends S5 June 2024 – August 2024
Buildspace *San Francisco, USA — Remote*

- Selected for an exclusive startup program backed by Y Combinator and a16z, focused on rapid product development of Stapes AI.
- Accelerated development of Stapes AI through community feedback, mentorship, and rapid prototyping.
- Applied skills in Python, Flutter, and embedded systems (ESP IDF) to develop and refine product offerings.

Technical Lead April 2024 – Present
GeeksForGeeks *New Delhi, India — On-site*

- Conducted comprehensive machine learning workshops for 80+ students, teaching essential ML and DL principles and applications, which enhanced participants' practical skills.
- Organized and led 10+ GeeksforGeeks (GFG) workshops across SRM, facilitating doubt clearance sessions that enhanced coding proficiency for over 200 students.
- Led a team of 8 technical staff and manage project timelines for various initiatives, applying experience in troubleshooting and server management.

Core Technical Team Member October 2022 – April 2024
GeeksForGeeks *New Delhi, India — On-site*

- Contributed as a Core Member of GFG, organizing the Phoenix Hackathon, where I achieved 1st place.
- Organized and led Hack-Innovate, a two-day hackathon during the tech fest, attracting over 300 participants and showcasing 50+ innovative solutions.

Research

A Name Is Not a Memory: Relational Binding in Continual Parametric Memory for Conversational Agents *(In Progress)* August 2026 – Present

- **Goal:** Develop **retrieval-free** long-term memory for conversational agents by encoding user-specific knowledge in persistent LoRA adapters on a frozen language model; evaluate how parametric memory compares with retrieval-based systems such as Recallr AI on retention, identity binding, continual updates, and cross-user interference; and explore a scalable multi-LoRA architecture inspired by Mixture-of-Experts routing.

Hallucination Dynamics in Small Language Models: A Large-Scale Empirical Study *(In Progress)* January 2026 – Present

- **Goal:** Determine how sampling temperature affects hallucination behavior in small language models across model scales, quantization levels, and factual domains.
- **Approach:** Collected over 10 million data samples from SLMs including Qwen2.5-0.5B/1.5B/3B/7B, Qwen3-1.7B/4B/8B, Llama-3.2-1B/3B, Gemma-3-4B, and Mistral-7B. Tested $T \in [0, 1]$ at $\Delta T = 0.1$ on HaluEval-QA and True/False across politics, geography, history, science, health, cities, companies, facts, and other domains.
- **Result:** Identified a non-monotonic scale-temperature relationship: sub-1.5B SLMs often collapsed into one-class prediction priors at low temperature, so added sampling entropy could reduce hallucination error by weakening systematic bias; near the 1.5B competence boundary, higher temperature instead amplified factual instability, especially under 4-bit quantization, while many 3B–8B models remained comparatively stable. Effects varied by topic and were concentrated on marginal knowledge, showing that hallucination risk depends on the interaction of scale, calibration, quantization, and domain rather than temperature alone.

Recallr AI: Scalable Architecture for Dynamic Long-Term Memory in Conversational AI Agents *(In Progress)* April 2025 – Present

- **Goal:** Build scalable long-term memory for conversational AI that preserves evolving user state across sessions without adding latency to real-time responses.

- **Approach:** Built a versioned knowledge graph through decoupled asynchronous curation and synchronous retrieval. Its decision logic classifies new evidence as redundant, novel, additive, temporally superseding, expired, or directly conflicting; dual timestamps separate event time from discussion time, immutable version chains preserve state evolution, and a Git-inspired human-in-the-loop protocol resolves contradictions. Auto-Recall routes queries across Low-Latency, Balanced, and Agentic retrieval with adaptive graph traversal.
- **Result:** Benchmarked Recallr on the LongMemEval Oracle tier. Agentic Recall achieved 98.9% accuracy—32.7 percentage points above the nearest competitor—with 100% on knowledge updates, 99.2% on temporal reasoning, and 98.5% on multi-session questions; Low-Latency Recall retained 92.6% accuracy at 396 ms P95, approximately 4.5× faster at P95 than the fastest competing system.

Achievements

Y Combinator's AI Startup School: Selected among 2,000 top CS students globally to attend YC's first-ever AI Startup School in San Francisco. (2025)

Google InnoSprint Hackathon: 1st runner up. (2023)

Phoenix Hackathon: Secured 1st position. (2022)

Code-A-Thon: 1st runner up. (2022)

Live-Project Competition: 2nd runner up. (2023)

SIH: 1st runner up at college level. (2023)

Certifications

Data Mining — *IIT, Kharagpur*
Neural Networks and Deep Learning — *DeepLearning.AI*
PyTorch for Deep Learning Bootcamp — *Udemy*

Education

SRM Institute of Science and Technology — May 2022 – July 2026
Bachelor of Technology in Computer Science, Specialization in AI and ML; CGPA: 8.5 — *Chennai, India*

Extracurricular

Course Instructor — *September 2024 – April 2026*
GeeksForGeeks

Assistant Course Instructor — *July 2024 – September 2024*
GeeksForGeeks